Air Quality Monitoring with Arduino

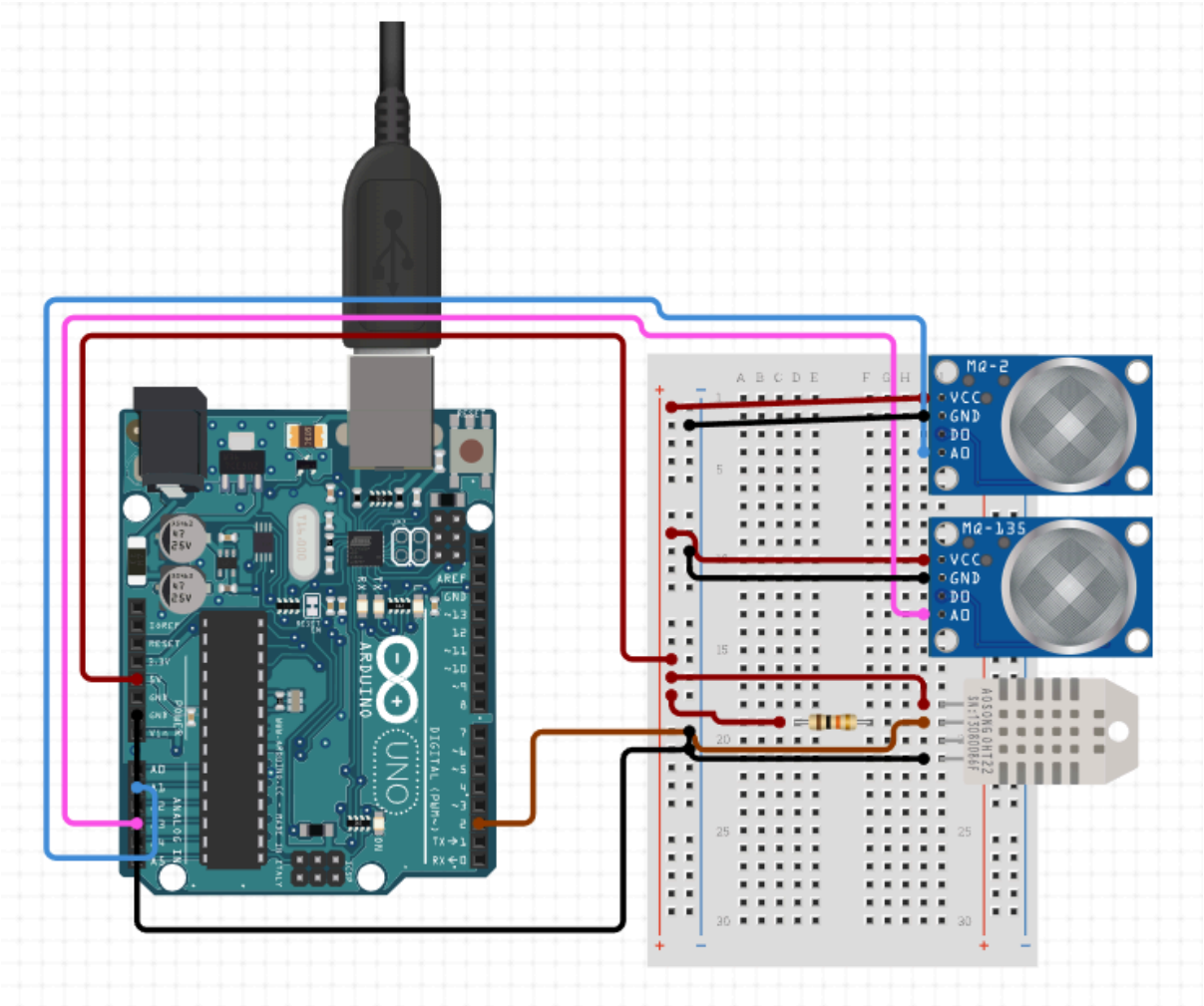
Introduction

This project presents a compact system for monitoring air quality using commonly available sensors. It features the MQ-2 and MQ-135 gas sensors for detecting various pollutants, combined with a DHT11 sensor for measuring temperature and humidity. The readings are converted into pollution levels expressed in both ppm and µg/m³, and corresponding air quality remarks are generated to provide easy interpretation of the data.

Bill of Materials (BoM)

Component	Quantity	Description
Arduino Uno	1	Microcontroller used for processing signals
MQ-2 Sensor	1	Detects smoke and carbon monoxide
MQ-135 Sensor	1	Detects harmful gases like NH_3, CO_2, etc.
DHT11 Sensor	1	Measures temperature and relative humidity
Breadboard	1	Used to assemble the circuit without soldering
Jumper Wires	As needed	For electrical connections between components
10kΩ Resistors	2	Load resistors for MQ sensor circuits

Circuit Diagram



Part 1 – Sensor Calibration (R_0 Calculation)

Purpose of R_0: Establish each sensor's baseline resistance in clean air. Future gas readings are compared against this value.

• Process Overview:

1. Take 500 analog readings from the sensor pin and compute their average.
2. Convert the average to voltage:
   voltage = (avgRaw × 5 V) / 1023.
3. Calculate sensor resistance (R□) in clean air:
   R□ = (Vcc × RL / voltage) − RL
   • Vcc = 5 V, RL = 10 kΩ.
4. Divide by the clean-air factor (4.4) to get R_0:
   R_0 = R□ / 4.4
   • Outcome: A stable R_0 for MQ-2 and MQ-135, recorded for use in pollution calculations.

Once these values stabilize, the calculated averages are stored for use in the next phase. *To improve stability, each new R_0 value is averaged with the previous value before being used, helping reduce noise and sudden spikes from the analog readings.*

```
Previous R0_MQ135 = 30.94
Current R0_MQ135 = 30.94
Average R0_MQ135 = 30.94
------ R0 Values ------
Previous R0_MQ2 = 12.24
Current R0_MQ2 = 12.24
Average R0_MQ2 = 12.24
Previous R0_MQ135 = 30.94
Current R0_MQ135 = 30.93
Average R0_MQ135 = 30.94
```

Part 2 – Air Pollution Measurement

Concentration Calculation (PPM & µg/m³)

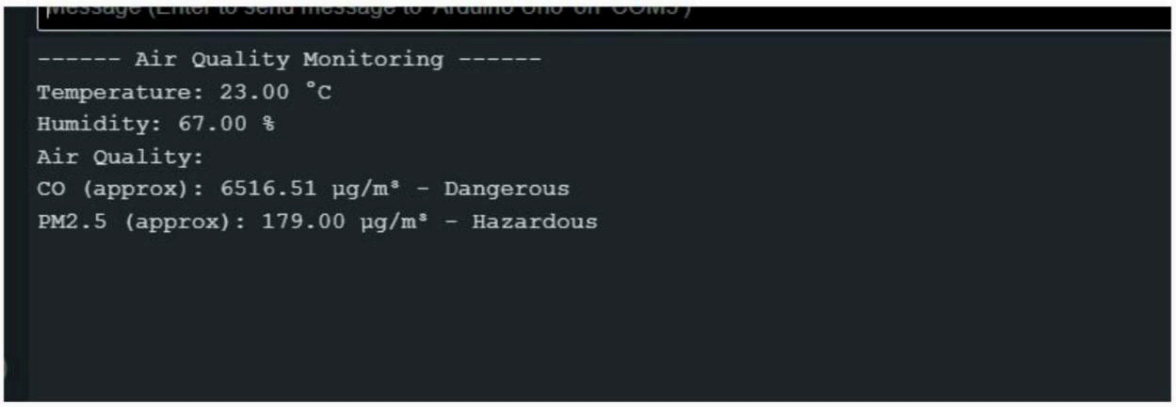
• Reading Rs in Polluted Air:
– Take a single analog reading, convert to voltage, then compute R□_gas as above.
• Compute Ratio:
ratio = R□_gas / R_0
• Convert Ratio to ppm (log–log curve):
ppm = 10^[($\log_{10}$(ratio) − b) / m]
– m (slope) and b (intercept) come from the sensor's datasheet.
• Convert ppm to µg/m³:
µg/m³ = ppm × (molecular weight / 24.45)
– 24.45 L/mol is the molar volume at 25 °C and 1 atm.

Each µg/m³ value is matched with a descriptive air quality level:
Good, *Moderate*, *Unhealthy*, *Hazardous*, etc.

```
Temperature: 25.50 °C
Humidity: 74.00 %
Air Quality:
MQ-2 (Smoke): 3366.29 µg/m³
MQ-135 (Air Quality): 216.50 µg/m³
```

Process Overview

1. Calibration is conducted using the R_0 calculation sketch, and stable average values are recorded.
2. These R_0 values are inserted into the monitoring code.
3. The system collects data, performs calculations, and outputs real-time air quality statistics via the Serial Monitor.



Optional Enhancements

- Displaying real-time data on an OLED or LCD screen.
- Logging measurements to an SD card for offline analysis.
- Uploading data to cloud platforms using modules such as ESP8266.
- Including additional environmental sensors for detecting NO_2, SO_2, or PM10.